

CLINOMA PLATFORM

FIRST PAPER CAMP

Questions & Model Answers Booklet

Day 1: Pediatrics

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CHAPTER 01

Growth and Development (Selected Topics)

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1. Short Essay & Enumerate (4 Points)

Q1: Enumerate four clinical uses of Percentile Curves in growth charts.

Model Answer:

- Early detection of protein energy malnutrition by following the flattening of the infant curve.
- Early detection of hydrocephalus by tracking the jumping of the infant curve to other above curves.
- Determination of risk for obesity if the body mass index curve lies between the 85th - 95th percentiles.
- Diagnosis of nutritional diseases by using Wellcome and Waterlow classifications.

Q2: Enumerate four key development warning signs in children.

Model Answer:

- Persistence of primitive reflexes > 6 months of age.
- Not walking by 18 months.
- No clear spoken words by 18 months.
- No two-word sentences by 2 years. (or: Discrepant head size or crossing centile lines / No response to environment or parent by 12 months).

2. Short Answered Questions (Definitions, Mention, List)

Q3: Define the terms "Growth" and "Development" according to your curriculum.

Answer:

- Growth: Natural increase in the size of the body either by (hyperplasia) through multiplication of different cells of different organs or (hypertrophy) through increase in the cell size.
- Development: Functional maturation of the central nervous system that leading to gaining skills and social adaptation.

Q4: List four major causes of Delayed Walking in pediatrics.

Answer:

- Cerebral palsy.
- Mental retardation.
- Peripheral nerves disorders.
- Muscles disorders.

3. Problem Solving (2 Cases)

Case 1:

A mother brings her 18-month-old male infant to the pediatric clinic because he is unable to walk alone or even stand with support. On physical examination, the infant shows a marked head lag when pulled up from a supine position, and an exaggerated "U-shape" is noted during ventral suspension.

Q1: What is your clinical definition or diagnosis for this presentation?

Answer:

Delayed walking due to severe persistent hypotonia (Floppy Infant syndrome).

Q2: Mention two lower motor unit levels or muscle disorders that can cause this presentation.

Answer:

1. Muscles disorders (such as Duchenne muscular dystrophy). 2. Peripheral nerves disorders.

Case 2:

During a routine health check-up for a 4-month-old infant, the pediatrician evaluates the developmental milestones. The infant is able to turn his head to sounds, laughs loudly, and can roll from back to front. However, on performing the neurological examination, the pediatrician notes that the Moro reflex is still fully persistent and can be easily elicited.

Q1: Based on the key development warning signs, why is this finding significant?

Answer:

Because the persistence of primitive reflexes (like the Moro reflex) beyond 6 months of age is considered a major development warning sign that requires serious evaluation.

Q2: List two gross motor and two social milestones that this infant should normally achieve at this exact age (4 months).

Answer:

- * Gross motor: Sits with support / Rolls from back to front.
- Social: Enjoys looking around / Recognize mother (or aware of strange faces).

CHAPTER 02

Nutrition & Nutritional Disorders (Selected Topics)

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1. Short Essay & Enumerate (4 Points)

Q1: Enumerate four anti-infective properties or factors found in human breast milk.

Model Answer:

- Secretory IgA: provides immunity against different pathogens like E.coli, shigella, salmonella, and viruses like poliovirus.
- Bifidus factor: stimulates the growth of lactobacillus bifidus which produces lactic acid, changing the pH to create an unsuitable medium for pathogens.
- Lactoferrin: an iron-binding protein that prevents iron uptake by organisms, exerting a bacteriostatic effect on E. coli.
- Lysozymes: enzymes that actively destroy the bacterial cell wall. (or: Phagocytes & lymphocytes / Complement / Interferon / B12 binding factor).

Q2: Enumerate the four constant features required to diagnose Kwashiorkor (KWO).

Model Answer:

- Growth failure: Failure to gain proper weight where weight for age is less than 80% of the reference.
- Pitting oedema: starts in the feet and lower legs, sometimes presenting as a puffy face with prominent cheeks (moon face).
- Mental changes: apathy, with no interest in surroundings and no smile.
- Muscle wasting with preserved subcutaneous fat.

Q3: Enumerate the classic radiological epiphyseal triad seen in an active wrist X-ray of a child with Rickets.

Model Answer:

- Broadening of the lower ends of the radius and ulna.
- Cupping (concave appearance of the metaphysis).
- Fraying (irregularity of the epiphyseal line).

2. Short Answered Questions (Definitions, Mention, List)

Q4: Mention the absolute maternal and infant contraindications to breastfeeding.

Answer:

- Absolute Maternal Contraindications: Cancer breast, Insanity of the mother, and HIV infection.
- Absolute Infant Contraindications: Inborn errors of metabolism, such as galactosemia.

Q5: Define Marasmus according to the Welcomme and anthropometric criteria.

Answer:

It is a severe form of protein energy malnutrition (PEM) characterized by a weight for age < 60% of the reference standard with completely absent edema, accompanied by gross loss of subcutaneous fat and obvious muscle wasting.

3. Problem Solving (2 Cases)

Case 1:

A 15-month-old female infant is brought to the clinic due to progressive failure to thrive and a change in her behavior. On examination, she is characteristically apathetic and never smiles. She has marked pitting edema in both feet and legs, and a prominent "moon face" appearance. Her subcutaneous fat seems relatively preserved, but her hair is dry, easily pickable, and shows alternating bands of normal coloration and hypopigmentation.

Q1: What is the accurate clinical diagnosis for this child?

Answer:

Kwashiorkor (KWO).

Q2: What is the medical term for the alternating hair hypopigmentation bands, and what does it signify?

Answer:

It is called the Flag sign. It signifies multiple relapses, representing alternating periods of a normal nutritional state and protein malnutrition.

Case 2:

A 10-month-old exclusively breastfed infant who lives in an apartment with no sun exposure is evaluated for delayed motor milestones. On examination, the infant has excessive frontal sweating and is highly irritable. Skeletal examination reveals an abnormal softness over the occipital bones, prominent widening of both wrists and ankles, and a visible palpable enlargement of the costochondral junctions on the chest wall.

Q1: What is your provisional diagnosis for this infant?

Answer:

Active Nutritional Rickets.

Q2: Based on your curriculum, state the medical term for the abnormal skull softness, and predict the expected serum levels of Calcium and Phosphorus in this active phase.

Answer:

- * Skull softness term: Craniotabes.
- Serum Calcium level: Normal (due to secondary hyperparathyroidism which mobilizes calcium from bone to blood).
- Serum Phosphorus level: Low / Decreased.

CHAPTER 03

G.I.T. (Selected Topics)

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1. Short Essay & Enumerate (4 Points)

Q1: Enumerate four clinical diagnostic Red Flags that suggest Organic Abdominal Pain in children.

Model Answer:

- Clearly localized pain (burning, stabbing, etc.) mostly away from the umbilicus.
- Pain awakening the child at night.
- Pain presenting with persistent fever or weight loss.
- Presence of blood in stools (either occult or obvious). (or: Tenderness/organomegaly - Anemia - Elevated ESR and CRP - Radiating pain).

Q2: Enumerate four acute clinical manifestations of IgE-Mediated Cow's Milk Allergy (CMA).

Model Answer:

- Skin signs: Urticaria, angioedema, flushing, or pruritus.
- Gastrointestinal signs: Vomiting, abdominal pain, or diarrhea.
- Respiratory signs: Wheeze, cough, stridor, or rhinitis.
- Systemic signs: Life-threatening Anaphylaxis.

2. Short Answered Questions (Definitions, Mention, List)

Q3: Define "The Acute Abdomen" and list its three key local physical signs.

Answer:

- Definition: A sudden onset of severe abdominal pain developing over a short time period.
- Three Key Local Physical Signs:
 - Guarding: contraction of abdominal muscles and discomfort when the doctor presses on the abdomen.
 - Rigidity: hardness of abdominal muscles.
 - Rebound tenderness: an increase in severe pain and discomfort when the doctor abruptly stops pressing on a localized region.

Q4: Differentiate accurately between the definitions of "Vomiting" and "Regurgitation".

Answer:

- Vomiting: Forceful oral expulsion of gastric contents associated with contraction of the abdominal and chest wall musculature.
- Regurgitation: The act by which food is brought back into the mouth without the abdominal and diaphragmatic muscular activity that characterizes vomiting.

3. Problem Solving (2 Cases)

Case 1:

An 8-year-old girl presents to the clinic with a history of recurrent episodes of abdominal pain over the past 4 months. The pain has a gradual onset and is characteristically localized in the periumbilical region with a poor relationship to food intake or defecation. She is completely symptom-free between episodes. On physical examination, her growth is normal, and there is no localized abdominal tenderness or organomegaly.

Q1: What is the most likely diagnosis for this child?

Answer:

Dysfunctional Abdominal Pain / Functional Abdominal Pain disorder.

Q2: Mention two associated autonomic symptoms that can be seen in about 30 percent of such cases.

Answer:

Pallor, nausea, fatigue, or anxiety.

Case 2:

A 3-month-old formula-fed male infant is brought to the outpatient clinic with chronic diarrhea, failure to thrive, and frequent streaks of blood and mucus in his stool. Physical examination reveals chronic, severe atopic dermatitis on his cheeks that is refractory to local treatment. The symptoms started 3 weeks after introducing standard cow's milk formula.

Q1: What is the clinical diagnosis for this infant?

Answer:

Non-IgE-mediated Cow's Milk Allergy (CMA) [Delayed type].

Q2: Mention the gold standard diagnostic test to confirm this condition, and state the first-line formula recommended for this formula-fed infant.

Answer:

- * Gold Standard Test: Oral Food Challenge performed under medical supervision.
- First-line Formula: Extensively Hydrolyzed Formula (eHF).

1. Short Essay & Enumerate (4 Points)

Q1: Enumerate four indications or limitations where Oral Rehydration Solution (ORS) is considered NOT suitable or has limitations in management.

Model Answer:

- Severe dehydration (requires urgent IV therapy).
- Paralytic ileus.
- Rapid stool losses ($>15 \text{ ml/kg/hr}$) with inability to drink fluid at a sufficient rate to replace losses.
- Glucose malabsorption (where ORS is not absorbed, stool volume increases, and dehydration worsens).

Q2: Enumerate four clinical diagnostic "Red Flags" that a mother should look for to immediately seek medical hospital evaluation for a child with diarrhea.

Model Answer:

- Many watery stools.
- Eating or drinking poorly.
- Repeated vomiting.
- Blood in the stools.
- (or: Becomes very thirsty - Fever - Does not get better in 3 days).

2. Short Answered Questions (Definitions, Mention, List)

Q3: Define "Dysentery" and differentiate between acute, persistent, and chronic diarrhea based on duration.

Answer:

- Dysentery: Diarrhea with visible blood in stool.
- Acute Diarrhea: Duration is less than 2 weeks.
- Persistent Diarrhea: Duration is more than 2 weeks.
- Chronic Diarrhea: Duration is more than 4 weeks.

Q4: List four medical (infective or non-infective) etiologies that can cause Acute Diarrhea in children.

Answer:

- Bacterial infection: e.g., Salmonella, E.coli, or Shigella.
- Viral infection: Rotavirus or Norovirus.
- Parasitic infection: Entamoeba histolytica or Giardia lamblia.
- Drug-induced: Antibiotics, especially oral ampicillin.
- (or: Non-infective dietetic causes like overfeeding or inappropriate foods).

3. Problem Solving (2 Cases)

Case 3:

A 10-month-old infant is brought to the dehydration center with a 2-day history of acute watery diarrhea and frequent vomiting. On examination, the infant is highly restless and irritable. His eyes appear characteristically sunken. When the mother offers water, the infant drinks it thirstily and eagerly. When the pediatrician performs a skin pinch over the abdomen, the skin turgor goes back slowly.

Q1: What is the accurate classification of dehydration for this infant, and what is the appropriate management plan?

Answer:

- * Classification: Some Dehydration (Fluid deficit 5-10% of body weight).
- Management Plan: Plan B in the Dehydration Centre (providing 75 ml/kg of ORS over 4-6 hours).

Q2: Based on your curriculum, how should the ORS be administered, and what should be done if the infant vomits during administration?

Answer:

- * Administration: Given by cup and spoon, 1 teaspoonful every 1-2 minutes.
- If vomiting occurs: Give the child a rest for 10 minutes and then restart again more slowly. If vomiting still continues >4 times in one hour, give ORS by a nasogastric tube (NGT).

Case 4:

A 5-week-old male infant presents with nonbilious projectile vomiting immediately after feeding. The mother states that the emesis has been progressive over the last week, and the infant is always intensely hungry and wants to feed again immediately after vomiting. On physical examination, the infant shows signs of severe dehydration, visible gastric peristalsis is noted over the abdomen, and an olive-shaped hard mass is clearly palpated above and to the right of the umbilicus.

Q1: What is the clinical diagnosis, and what is the diagnostic thickness or length criteria required to confirm it on an Ultrasound examination?

Answer:

- * Diagnosis: Hypertrophic Pyloric Stenosis (HPS).
- Ultrasound Criteria: Pyloric thickness greater than 4 mm or an overall pyloric length greater than 14 mm .

Q2: State the expected laboratory acid-base disturbance for this condition, and mention the surgical procedure of choice.

Answer:

- * Laboratory finding: Hypochloremic metabolic alkalosis.
- Surgical Procedure: Ramstedt pyloromyotomy.

CHAPTER 04

Genetic Diseases (Selected Topics)

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1. Short Essay & Enumerate (4 Points)

Q1: Enumerate four dysmorphic features seen in the face and head of a child with Down Syndrome.

Model Answer:

- Upward slanting palpebral fissures.
- Epicanthus and Brushfield spots of the iris.
- Small dysplastic pinnae and low-set ears.
- Mid-face hypoplasia, flat nasal bridge, and protruded tongue.

Q2: Enumerate four clinical indications for performing Prenatal Diagnosis in a pregnant woman.

Model Answer:

- Advanced maternal age where the pregnant woman is aged 35 years or more.
- History of a previous child with chromosomal abnormalities or inherited disorders.
- History of recurrent abortions in the family or parents.
- Parents who are known carriers for balanced chromosomal arrangements or X-linked inherited disorders.

2. Short Answered Questions (Definitions, Mention, List)

Q3: Define Turner Syndrome and mention its most common genotype.

Answer:

- Definition: A chromosomal disorder affecting females characterized by severe short stature, short webbed neck, low posterior hairline, lack of development of secondary sexual characteristics due to gonadal dysgenesis, and primary amenorrhea.
- Most Common Genotype: 45,X.

Q4: List four invasive or non-invasive types/methods used for Prenatal Diagnosis.

Answer:

- Ultrasonography (Non-invasive).
- Chorionic villus sampling (CVS) (Invasive).
- Amniocentesis (Invasive).
- Fetal blood sampling or Cordocentesis (Invasive).

3. Problem Solving (2 Cases)

Case 1:

A 41-year-old pregnant female gives birth to a male newborn. On physical examination, the neonate shows generalized hypotonia, a flat occiput, upward slanting palpebral fissures, a flat midface with a protruding tongue, and a single transverse palmar crease on both hands. Auscultation of the heart reveals a loud pansystolic murmur at the lower left sternal border.

Q1: What is your provisional diagnosis for this neonate?

Answer:

Down Syndrome (Trisomy 21) associated with a Congenital Heart Defect.

Q2: Mention the most common cytogenetic type responsible for this condition and state its recurrence risk.

Answer:

Nondisjunction type (occurs in 94% of cases). It is a sporadic event and has a very low recurrence risk.

Case 2:

A 14-year-old girl is brought to the pediatric endocrine clinic due to failure to achieve menarche and severe short stature. On examination, her intelligence is completely normal, but she has a short webbed neck, a broad chest with widely spaced nipples, and a wide carrying angle at her elbows. Her blood pressure is persistently elevated at 140/90 mmHg.

Q1: What is the most likely clinical diagnosis?

Answer:

Turner Syndrome.

Q2: Based on your curriculum, explain the importance of prenatal screening for this disorder using the Maternal Serum Triple Test.

Answer:

The importance lies in early prenatal detection, which is characterized by a specific triple screen pattern showing normal or decreased AFP, decreased uE3, and characteristically increased HCG.